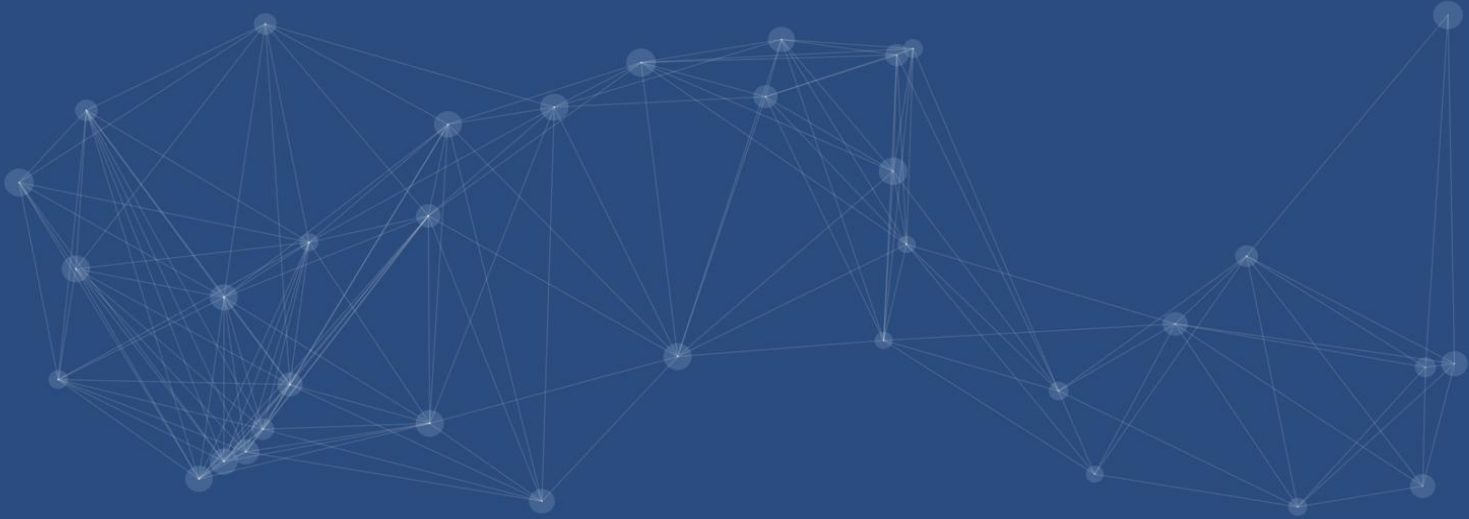


Investigating Gender Bias in Machine Learning

A Fairness Audit of Income Classification

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Artificial Intelligence Ethics and Their Applications

Word Count: 2,553



1. Introduction

Machine learning models are increasingly deployed in high stakes domains such as hiring, credit scoring, and healthcare, yet growing evidence demonstrates that these systems can encode and amplify societal biases. Algorithms trained on historically biased data frequently inherit the prejudices of prior decision makers, producing outcomes that disproportionately disadvantage protected groups even without any explicit discriminatory intent. In income prediction specifically, models trained on census data have been shown to systematically underestimate the earning potential of women, reflecting historical gender pay gaps embedded in the training data. The real world consequences of such bias are significant. A widely used healthcare algorithm was found to exhibit racial bias affecting millions of patients, reducing the number of Black patients identified for additional care by more than half. In financial contexts, biased income classifiers could lead to qualified women being denied loans, credit, or employment opportunities, thereby perpetuating cycles of economic inequality.

This report investigates gender bias in a **Random Forest** classifier trained on the **UCI Adult Census Income** dataset to predict whether an individual earns above or below \$50,000 per year. Sex is selected as the protected characteristic. The report summarises the data exploration and pre processing pipeline, evaluates overall model performance, and applies three fairness criteria, **equal accuracy**, **demographic parity**, and **equal opportunity**, to assess whether the model treats male and female groups equitably.

2. Model Development and Application of Fairness Criteria

2.1 Data Exploration

The **UCI Adult Census Income** dataset, originally extracted from the 1994 United States Census Bureau database, was used for this investigation. The dataset contains 32,537 records and 15 features, including six numerical columns (age, fnlwgt, education num, capital gain, capital loss, hours per week), eight categorical columns (workclass, education, marital status, occupation, relationship, race, sex, native country), and one binary target variable (income), where 0 represents an annual income of $\leq 50K$ and 1 represents $> 50K$. Exploratory data analysis was conducted following the four pillars of EDA.

Descriptive Statistics

Descriptive statistics revealed that **capital gain** and **capital loss** exhibited substantial differences between their mean and median values, indicating heavy right skew, with the

majority of individuals recording zero for both features. The income distribution confirmed significant class imbalance, with 24,698 records (75.9%) in the $\leq 50K$ class and only 7,839 records (24.1%) in the $>50K$ class.

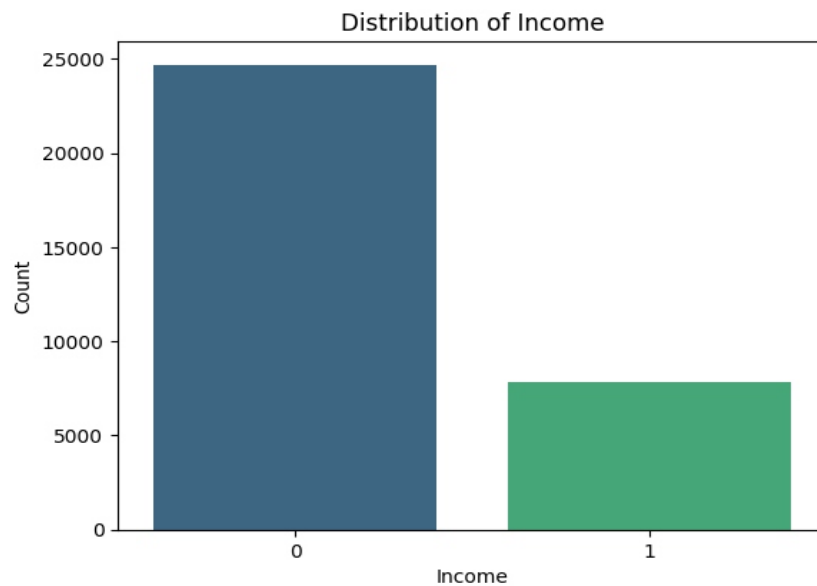


Figure 1: Distribution of Income Classes

Data Visualisation

Histograms highlighted that **age** was approximately normally distributed, while **capital gain** and **capital loss** were heavily zero inflated. The distribution of numerical features across the dataset is shown below.

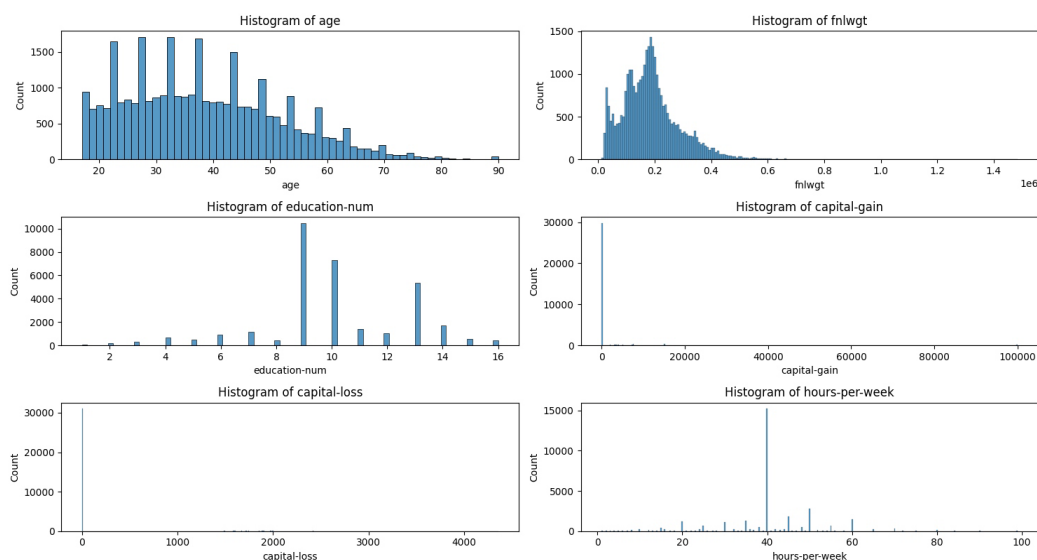


Figure 2: Histograms of Numerical Features

Boxplots further confirmed the presence of extreme outliers in **capital gain** and **capital loss**, while **age** and **hours per week** displayed more symmetric distributions with moderate outlier counts.

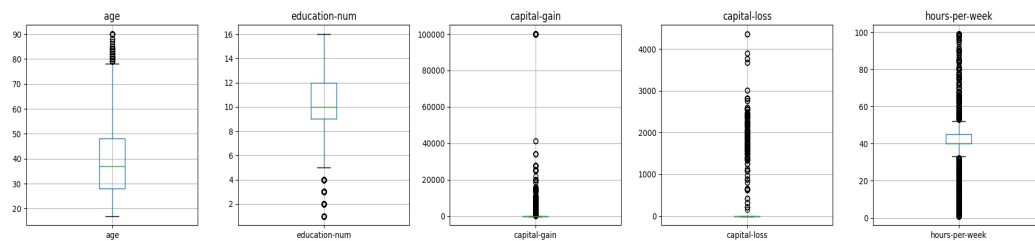


Figure 3: Boxplots of Numerical Features

Categorical analysis revealed that the dataset was predominantly composed of individuals from the United States, employed in the private sector, with high school or bachelor level education. Critically for this fairness investigation, the sex distribution showed a 2:1 male to female ratio, and among those earning >50K, approximately 30.6% were male compared to only 10.9% female, highlighting a pre existing representation disparity that the model could inherit.

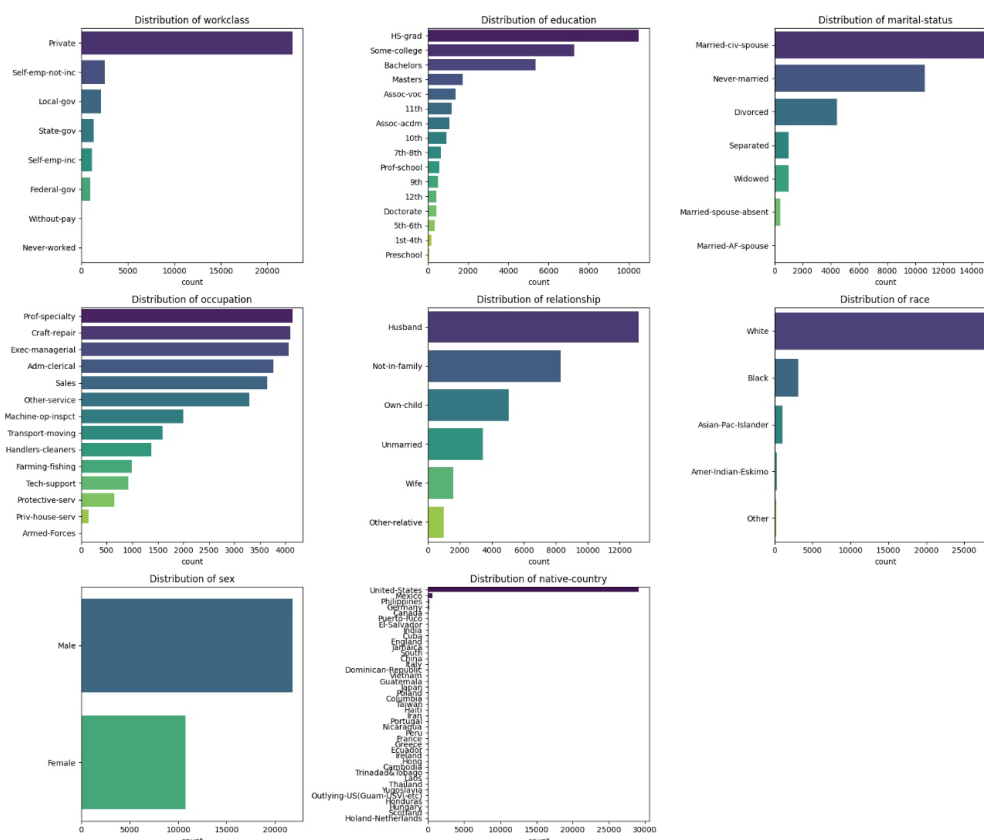


Figure 4: Distribution of Categorical Features

Correlation Analysis

Pearson correlation heatmaps were computed to identify linear relationships between features. The **fnlwgt** feature showed negligible correlation with all other variables, while **education num** displayed the strongest positive association with income.

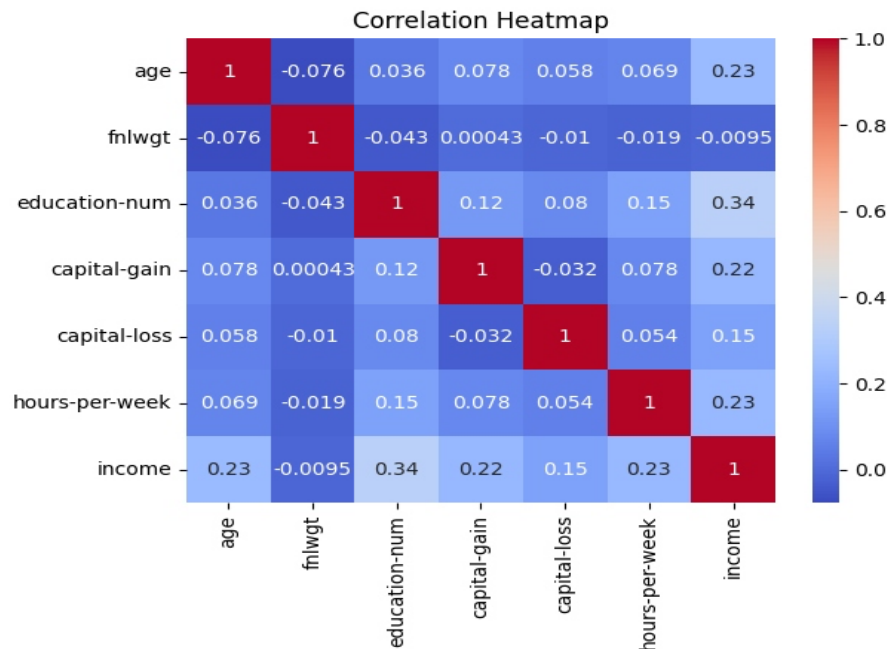


Figure 5: Pearson Correlation Heatmap

Spearman correlation analysis was also performed to capture monotonic relationships. The results confirmed the same patterns observed in the Pearson analysis, with **education num** and **age** showing the strongest associations with the target variable.

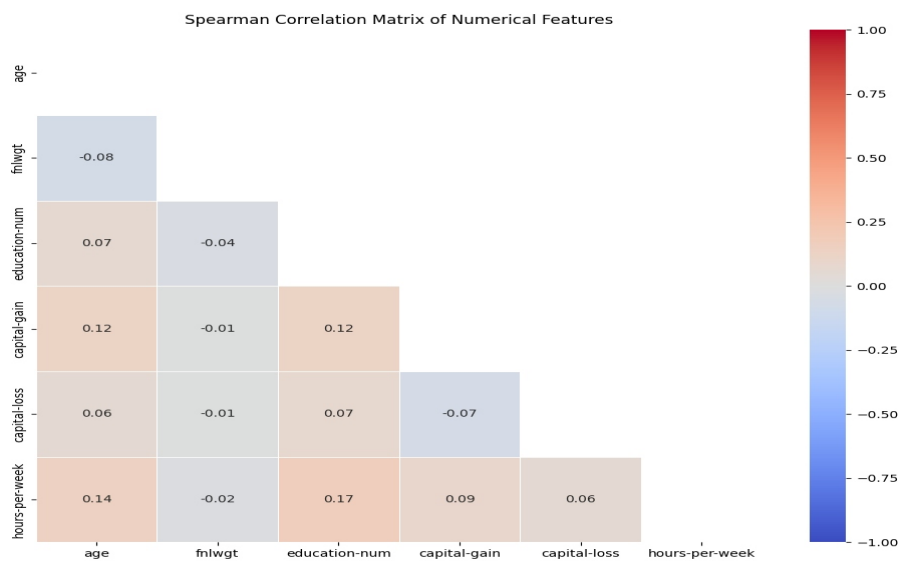


Figure 6: Spearman Correlation Heatmap

Distribution Assessment

Kernel density estimation plots and **Quantile Quantile** plots confirmed that most numerical features deviated from normality, with **capital gain** and **capital loss** showing particularly extreme departures due to their zero inflated distributions.

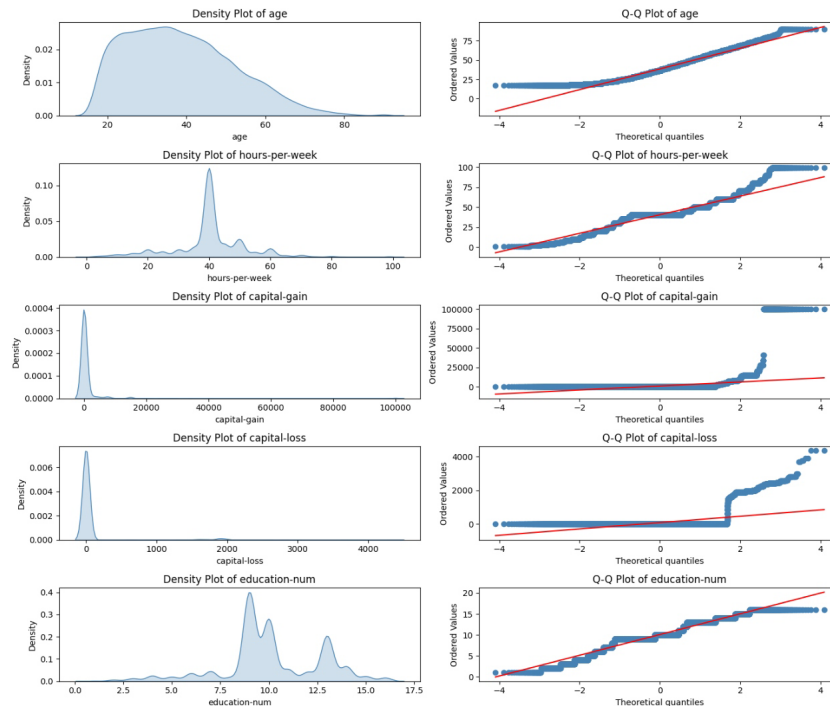


Figure 7: Density Plots and Q Q Plots

2.2 Data Pre processing

Data cleaning began with feature removal. The **fnlwgt** column was dropped because correlation analysis showed it had no meaningful relationship with the target variable or any other feature. The **education** column was also removed because its numerical equivalent, **education num**, already captured the same information as a correctly ordered ordinal variable, thus avoiding redundancy without losing predictive value.

Data types were then corrected to ensure categorical columns were stored as string types and numerical columns as integers, enabling the pipeline to apply appropriate transformations to each group. Missing values were identified in three categorical columns: **occupation** (5.66%), **workclass** (5.64%), and **native country** (1.79%). Since all missing values occurred in categorical variables and represented less than 6% of the total data, **mode imputation** was applied. Although advanced imputation techniques such as **KNN** and **MICE** exist, they were not appropriate for this dataset. Simple mode imputation is statistically robust for small proportions of categorical missingness, computationally efficient, and avoids introducing unnecessary model complexity that could itself bias the fairness audit. Duplicate rows were subsequently removed.

Outlier analysis was performed using **Z scores** on numerical features. While **capital gain** and **capital loss** contained extreme values, these were retained as legitimate data points reflecting real socio economic variation rather than measurement errors.

A **scikit learn ColumnTransformer** pipeline was constructed to handle all remaining transformations. **MinMaxScaler** was applied to numerical features to normalise values to a 0 to 1 range. **One Hot Encoding** was selected for all categorical variables to preserve interpretability and transparency, which is essential for a fairness focused audit. Alternative encoders such as **frequency encoding**, **binary encoding**, and **ordinal encoding** were intentionally avoided because they can collapse distinct categories, obscure the meaning of sensitive attributes, or introduce artificial ordinal relationships where none exist.

2.3 Model Development

The cleaned dataset was split into training (80%) and testing (20%) sets using a **stratified train test split** with a fixed random state of 42. Stratification on the target variable was essential because the dataset exhibits significant class imbalance, with approximately 75% of records belonging to the $\leq 50K$ class. Without stratification, the minority class could be under or over represented in either split, leading to unreliable performance estimates and masking potential fairness issues. A copy of the test set features (**X_test_audit**) was preserved prior to transformation to retain the original sex column for the subsequent fairness audit.

A **Random Forest** classifier was selected as the modelling algorithm, configured with 100 estimators and the **class_weight** parameter set to **balanced**. **Random Forest** is an ensemble method that constructs multiple decision trees during training and aggregates their predictions, which makes it robust to overfitting and capable of capturing complex non linear relationships within socio economic data. The **balanced** class weight setting automatically adjusts sample weights inversely proportional to class frequencies, assigning higher importance to the under represented $> 50K$ class during training. This was a deliberate choice to mitigate the risk of the model defaulting to majority class predictions, which would inflate accuracy while concealing poor performance on the minority class.

The full modelling workflow was encapsulated within a **scikit learn Pipeline**, chaining the **ColumnTransformer** (containing **MinMaxScaler** and **OneHotEncoder**) with the **Random Forest** classifier. This ensures that all pre processing transformations are applied consistently to both training and test data, preventing data leakage.

2.4 Performance Evaluation

The trained **Random Forest** model was evaluated on the held out test set (5,794 records) using multiple complementary metrics. Table 1 summarises the overall results.

Table 1: Overall Model Performance

Metric	Value
Accuracy	0.84
ROC AUC	0.8813
F1 Score ($\leq 50K$)	0.89
F1 Score ($> 50K$)	0.65

The model achieved an overall accuracy of 84% and a **ROC AUC** of 0.8813, indicating strong discriminative ability in distinguishing between the two income classes. The **ROC curve** confirms that the model performs substantially better than a random baseline across all classification thresholds.

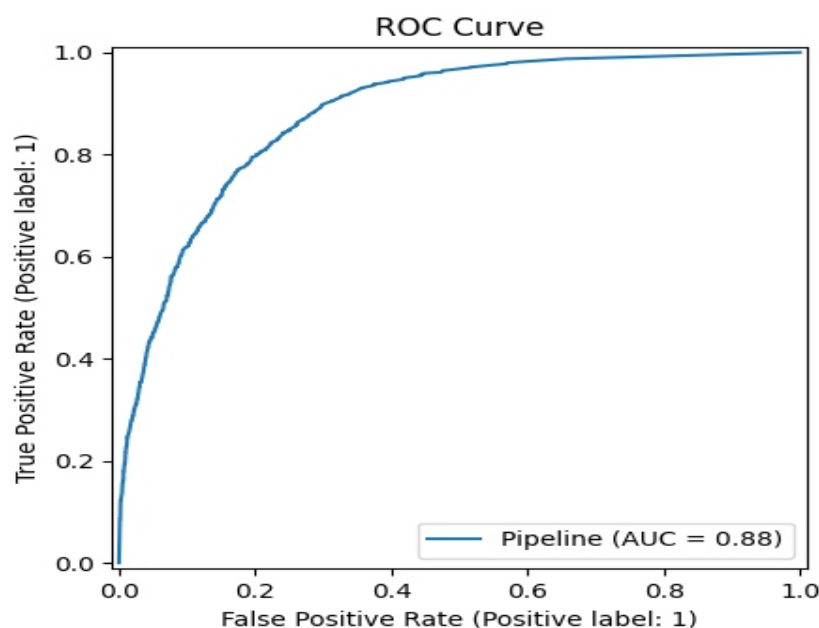


Figure 8: ROC Curve (AUC = 0.8813)

However, a notable discrepancy exists between the **F1 scores** for the two classes. The majority class ($\leq 50K$) achieves an F1 of 0.89, while the minority class ($> 50K$) achieves only 0.65. This gap indicates that the model struggles to correctly identify high income individuals, with precision and recall both lower for this group. The disparity is primarily explained by the class imbalance in the dataset, where high income individuals constitute only 25% of records.

The confusion matrix provides further detail. Of the 4,353 individuals earning $\leq 50K$, the model correctly classified 3,947 (**true negatives**), while misclassifying 406 as high income (**false**

positives). Of the 1,441 individuals earning >50K, the model correctly identified 880 (**true positives**) but missed 561 (**false negatives**).

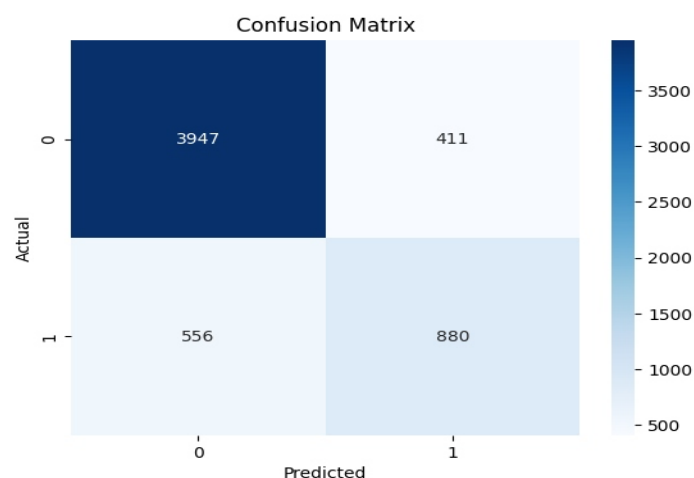


Figure 9: Confusion Matrix

The percentage confusion matrix further illustrates that 38.9% of genuinely high income individuals were incorrectly classified as low income. While the model demonstrates strong aggregate performance, an overall accuracy figure alone is insufficient for assessing fairness, as it may conceal disparate error rates across demographic groups. The following sections investigate this directly.

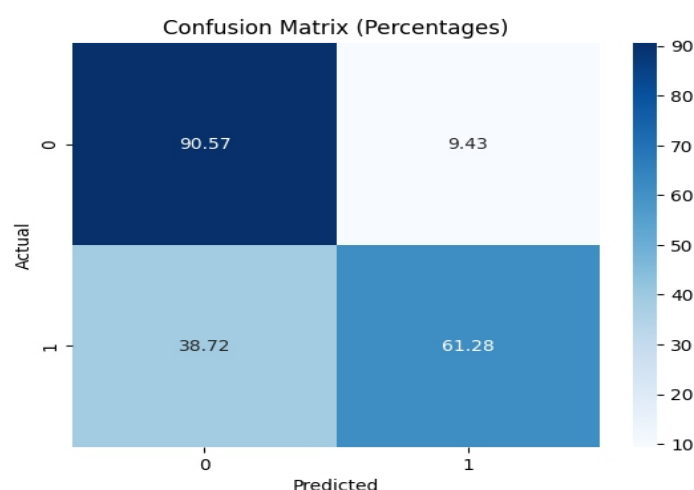


Figure 10: Confusion Matrix (Percentages)

2.5 Fairness Audit

To assess fairness, the test dataset was split into two groups based on the protected characteristic sex: a **male group** (n = 3,814) and a **female group** (n = 1,980). Separate confusion matrices and classification reports were generated for each group. Three fairness criteria were then applied. In fairness auditing practice, a common threshold of 5% is used: if

the absolute difference between the two groups on any criterion exceeds this value, the model is considered to exhibit bias against the disadvantaged group.

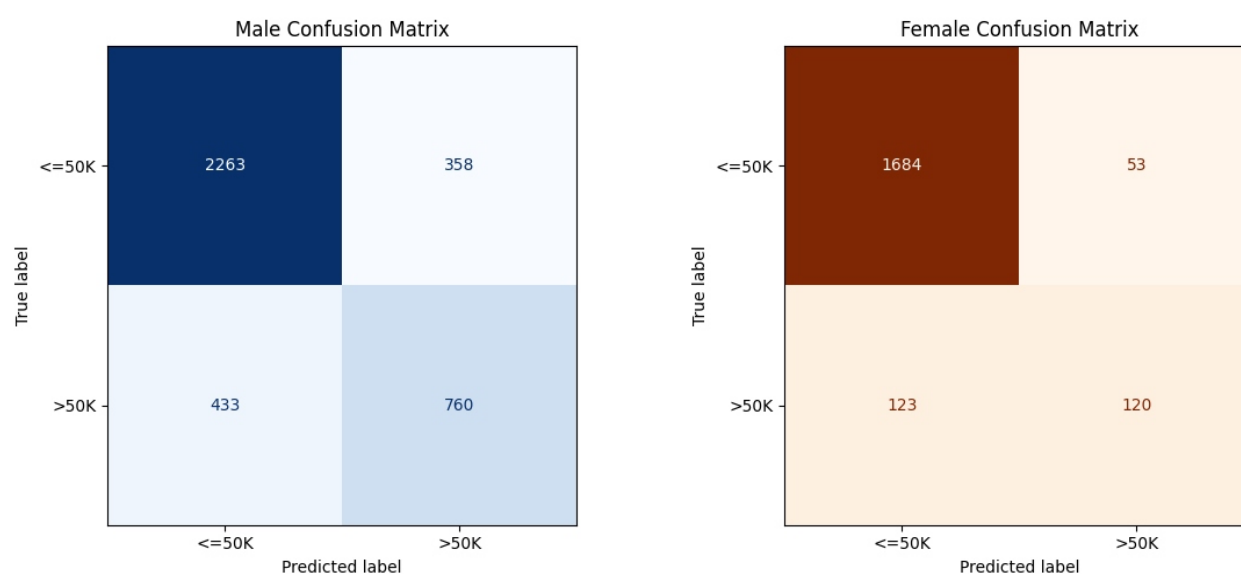


Figure 11: Confusion Matrices by Sex (Male and Female)

The matrix counts used to compute all three fairness criteria are summarised below. For the male group: **TP** = 760, **FP** = 358, **FN** = 433, **TN** = 2,263. For the female group: **TP** = 120, **FP** = 53, **FN** = 123, **TN** = 1,684.

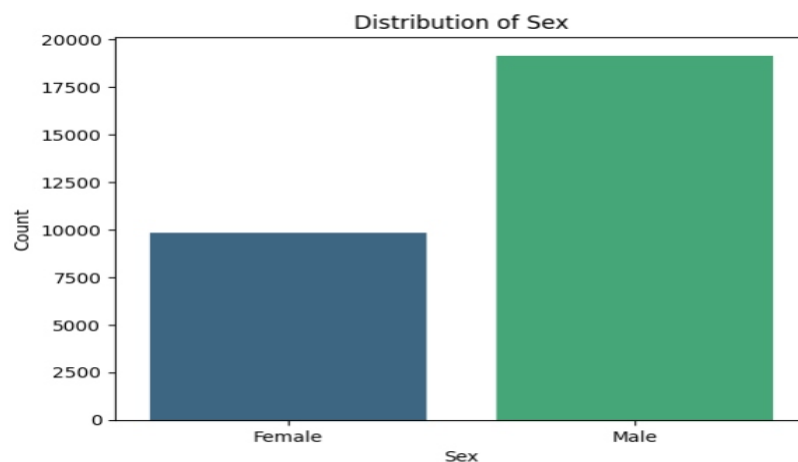
```

--- MATRIX COUNTS FOR FAIRNESS CRITERIA ---
MALE Matrix:  TP=760, FP=358, FN=433, TN=2263
FEMALE Matrix: TP=120, FP=53, FN=123, TN=1684

```

The **Equality Act 2010** identifies nine protected characteristics, including age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex, and sexual orientation. Sex was chosen for this audit because the dataset contains a clear sex attribute with a pronounced imbalance: approximately two thirds of

records are male and only one third female. This imbalance, combined with the well documented historical gender pay gap reflected in the 1994 census data, makes sex the most salient characteristic for investigating whether the classifier reproduces or amplifies structural inequality. The distribution of sex in the dataset is shown below.



2.5.1 Equal Accuracy

Equal accuracy requires that the model achieves the same overall accuracy for both demographic groups. The female group achieved 91.11% accuracy compared to 79.26% for the male group, producing a gap of 11.85% which exceeds the 5% threshold. **Bias is detected under this criterion.** However, this result must be interpreted with caution. The female group contains a much smaller proportion of high income individuals, with only 12.3% of females in the test set earning >50K compared to 31.3% of males. In such an imbalanced setting, a model can achieve high accuracy simply by predicting the majority class ($\leq 50K$) for nearly all female records. This phenomenon is known as the **accuracy paradox**, and it demonstrates that equal accuracy alone can be a misleading indicator of fair treatment.

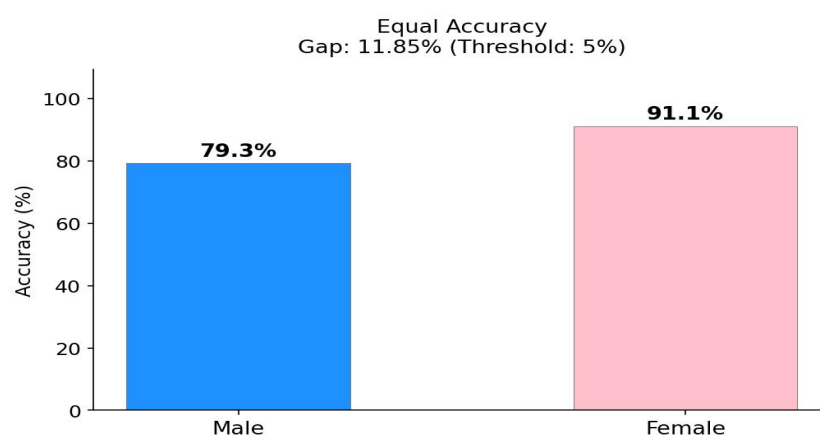


Figure 12: Equal Accuracy Comparison by Sex

CODE SNAP WITH FORMULA:

```
EQUAL ACCURACY

male_accuracy = (male_TP + male_TN) / (male_TP + male_TN + male_FP + male_FN)
female_accuracy = (female_TP + female_TN) / (female_TP + female_TN + female_FP + female_FN)
diff_accuracy = abs(male_accuracy - female_accuracy)

print("\n1. EQUAL ACCURACY")
print(f" Male Accuracy : {male_accuracy * 100:.2f}%")
print(f" Female Accuracy : {female_accuracy * 100:.2f}%")
print(f" Difference : {diff_accuracy * 100:.2f}%")
if diff_accuracy > 0.10:
    print(" BIAS DETECTED ")
else:
    print(" No significant bias ")

0.0s

1. EQUAL ACCURACY
Male Accuracy : 79.26%
Female Accuracy : 91.11%
Difference : 11.85%
BIAS DETECTED
```

2.5.2 Demographic Parity

Demographic parity requires that the model predicts the positive outcome (>50K) at the same rate for both groups, regardless of the actual ground truth labels. The model predicts high income for males at more than three times the rate it does for females (29.31% versus 8.74%), producing a gap of 20.58%, the largest disparity observed across all three criteria.

Bias is detected under this criterion. This result indicates that the model has learned to associate male demographic features with higher income far more readily than female ones. If such a model were deployed in a real world context such as loan approvals or hiring, women would be systematically less likely to receive favourable outcomes compared to equally situated men, constituting direct allocative harm. However, demographic parity does not account for base rate differences between groups. In the 1994 census data, women genuinely earned less than men due to historical labour market discrimination. Therefore, this criterion cannot distinguish between bias introduced by the model and inequality already present in the real world.

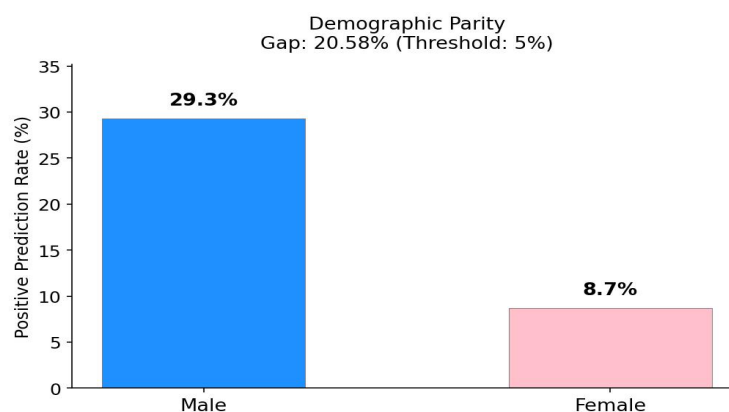


Figure 13: Demographic Parity Comparison by Sex

CODE SNAP WITH FORMULA:

```
DEMOGRAPHIC PARITY

male_positive_rate = (male_TP + male_FP) / (male_TP + male_TN + male_FP + male_FN)
female_positive_rate = (female_TP + female_FP) / (female_TP + female_TN + female_FP + female_FN)

disparate_impact_ratio = female_positive_rate / (male_positive_rate)

print(f"\n2. DEMOGRAPHIC PARITY")
print(f" Male predicted rich rate : {male_positive_rate * 100:.2f}%")
print(f" Female predicted rich rate : {female_positive_rate * 100:.2f}%")
print(f" Disparate Impact Ratio : {disparate_impact_ratio * 100:.1f}%")
if disparate_impact_ratio < 0.80:
    print(" BIAS DETECTED (Fails Four-Fifths Rule) ")
else:
    print(" No significant bias (Passes Four-Fifths Rule) ")

✓ 0.0s

2. DEMOGRAPHIC PARITY
Male predicted rich rate : 29.31%
Female predicted rich rate : 8.74%
Disparate Impact Ratio : 29.8%
BIAS DETECTED (Fails Four-Fifths Rule)
```

2.5.3 Equal Opportunity

Equal opportunity requires that the model correctly identifies positive class individuals at the same rate for both groups, measured by recall. The model correctly identifies high income males 63.70% of the time, but only 49.38% of high income females, producing a gap of 14.32%. **Bias is detected under this criterion.** This is arguably the most critical ethical failure revealed by the audit. Even when a woman has the exact same ground truth high income status as a man, the model is substantially less likely to recognise her as such. In practical terms, roughly one in two genuinely high income women are misclassified as low income, compared to approximately one in three high income men. If deployed in contexts such as credit scoring or automated hiring, this disparity would mean that qualified women are disproportionately rejected compared to equally qualified men, representing direct allocative harm. The **equal opportunity** violation is particularly concerning because, unlike equal accuracy or demographic parity, it conditions on the true label, meaning it captures unfairness that cannot be explained away by differing base rates alone.

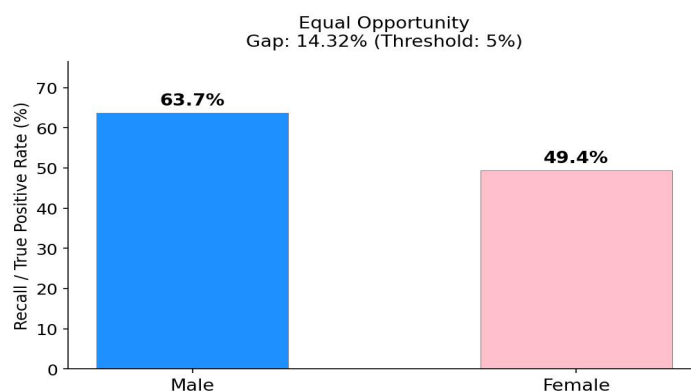


Figure 14: Equal Opportunity Comparison by Sex

CODE SNAP WITH FORMULA:

EQUAL OPPORTUNITY

```
male_recall = male_TP / (male_TP + male_FN)
female_recall = female_TP / (female_TP + female_FN)
diff_recall = abs(male_recall - female_recall)

print(f"\n3. EQUAL OPPORTUNITY")
print(f"   Male Recall (rich) : {male_recall * 100:.2f}%")
print(f"   Female Recall (rich) : {female_recall * 100:.2f}%")
print(f"   Difference : {diff_recall * 100:.2f}%")
if diff_recall > 0.05:
    print("   BIAS DETECTED ")
else:
    print("   No significant bias ")
```

87] ✓ 0.0s

```
3. EQUAL OPPORTUNITY
Male Recall (rich) : 63.70%
Female Recall (rich) : 49.38%
Difference : 14.32%
BIAS DETECTED
```

Fairness Audit Results

Table 2 consolidates the results of all three fairness criteria. All three criteria exceeded the 5% threshold, confirming systematic gender bias across every dimension of fairness assessed.

Table 2: Fairness Audit Results

Criterion	Male	Female	Gap	Result
Equal Accuracy	79.26%	91.11%	11.85%	Bias Detected
Demographic Parity	29.31%	8.74%	20.58%	Bias Detected
Equal Opportunity	63.70%	49.38%	14.32%	Bias Detected

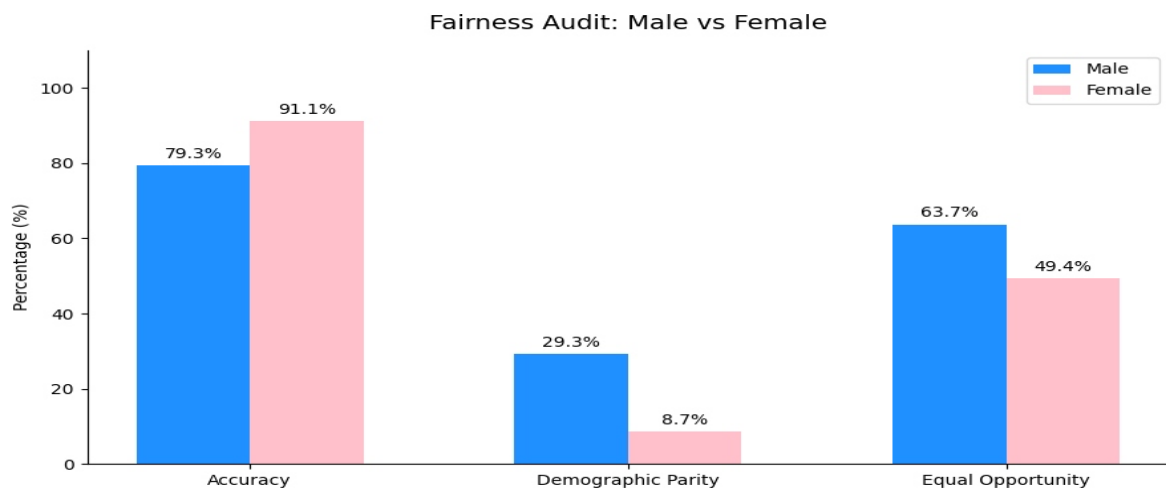


Figure 15: Fairness Audit Comparison — Male vs Female

3. Findings

The fairness audit confirmed the existence of systematic gender bias in the **Random Forest** income classifier across all three criteria. **Equal accuracy** revealed an 11.85% gap, **demographic parity** a 20.58% gap, and **equal opportunity** a 14.32% gap, each substantially exceeding the 5% threshold. The model consistently disadvantages women in high income prediction, with the most severe violation occurring in **demographic parity**, where males are predicted as high income at more than three times the rate of females.

However, each fairness criterion carries inherent limitations that must be acknowledged. **Equal accuracy** is susceptible to the **accuracy paradox**: when class distributions differ sharply between groups, a model can achieve artificially high accuracy by defaulting to majority class predictions. In this case, the female group's 91.11% accuracy is misleading because only 12.3% of females in the test set are genuinely high income. **Demographic parity** cannot distinguish between bias introduced by the model and real world historical inequality. In the 1994 census data, women genuinely earned less due to structural labour market discrimination, so enforcing equal prediction rates could result in predicting high income for individuals who do not actually earn it. **Equal opportunity**, while more robust because it conditions on the true label, cannot be satisfied simultaneously with demographic parity when base rates differ between groups. This is known as the **fairness impossibility theorem**, which demonstrates that these criteria are mathematically incompatible unless both groups share identical base rates. No single fairness criterion is therefore sufficient, and auditors must evaluate multiple criteria in combination.

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Appendix: Model Card

Model Card — Income Classification Model, Version 1.0 — 2026

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1. Model Details

Name: Random Forest Income Classifier

Version: 1.0

Model Type: Binary Classification

Algorithm: Random Forest (n_estimators=100, class_weight=balanced)

Framework: Scikit learn

2. Intended Use

Primary Use: Academic research to investigate bias in machine learning models.

Intended Users: Researchers and students studying AI fairness.

Out of Scope: Real world financial decisions, loan or credit approval, hiring or HR screening, any production deployment affecting real people.

3. Factors

Demographic Factors: Sex (Male/Female), Race, Age, Nationality.

Instrumentation Factors: Quality of self reported census data; missing value density in Occupation and Workclass columns.

4. Metrics

Performance Metrics: Accuracy, F1 Score, ROC AUC, Recall, Precision.

Fairness Metrics: Equal Accuracy, Demographic Parity, Equal Opportunity.

5. Evaluation Data

Dataset: UCI Adult Census Income Dataset. **Size:** 5,794 records (20% stratified test split, random_state=42). **Class Distribution:** ≤50K: 4,353 (75%) | >50K: 1,441 (25%). **Sex Distribution:** Male: 3,814 (65%) | Female: 1,980 (35%).

6. Training Data

Dataset: 80% of adult_cleaned.csv (23,176 records). **Pre processing:** Removed fnlwgt and education columns; mode imputation for missing categorical values; **MinMaxScaler** on

numerical features; **OneHotEncoder** on categorical features; duplicates removed; outliers retained as real data points.

Known Biases: Historical 1994 gender pay gap; predominantly US population; underrepresentation of women in high income bracket.

7. Quantitative Analyses

Overall Performance:

Metric	Value
Accuracy	0.84
ROC AUC	0.8813
F1 ($\leq 50K$)	0.89
F1 ($> 50K$)	0.65

Group Level Performance:

Metric	Acc.	Prec.	Recall	F1	TP/FP	FN/TN
Male (3,814)	79.3%	0.68	0.64	0.66	760/358	433/2263
Female (1,980)	91.1%	0.69	0.49	0.58	120/53	123/1684

Fairness Results:

Criterion	Male	Female	Gap	Result
Equal Accuracy	79.3%	91.1%	11.85%	Bias Detected
Demographic Parity	29.3%	8.7%	20.58%	Bias Detected
Equal Opportunity	63.7%	49.4%	14.32%	Bias Detected

8. Ethical Considerations

Bias Finding: Systematic bias against women across all three fairness criteria; most significant violation in **Demographic Parity** (20.58% gap).

Groups at Risk: Women in high income brackets systematically under identified; minority racial groups may face intersectional bias.

Potential Real World Harms: Loan approvals (rich women denied credit unfairly), hiring systems (high earning women filtered out), credit scoring (women underestimated financially).

Historical Bias: Model trained on 1994 data with significant gender pay gap; training on this data risks perpetuating discrimination.

9. Caveats and Recommendations

Data is from 1994 and does not reflect current income patterns. Limited primarily to US population. Intersectional bias (e.g., race and sex combined) not fully explored. **Equal Accuracy** is misleading on imbalanced datasets. The **fairness impossibility theorem** means all three criteria cannot be satisfied simultaneously when base rates differ between groups.

DO NOT DEPLOY IN ANY REAL WORLD APPLICATION

THIS MODEL CARD IS FOR ACADEMIC USE ONLY